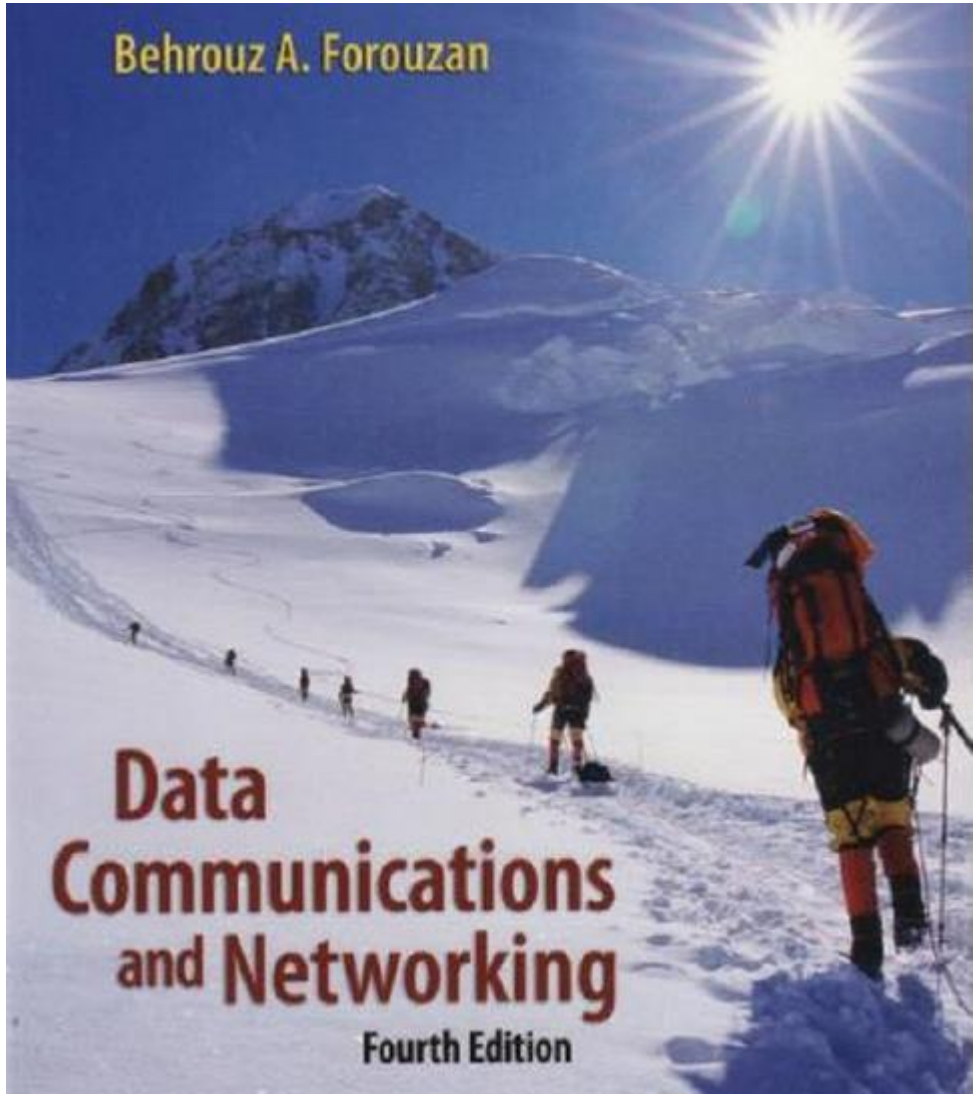
Chapter 19
Network Layer:
LOGICAL ADDRESSING



Lecture 16

Network Layer: Internet

- Network layer is concerned with getting packets from the source **all the way** to the destination (**end-end to transmission)**
- May require many hops at intermediate routers (**multiple hops**),
  - ➢ Its primary function is **routing**

IP addressing (IPv4) : introduction

- IP address: 32-bit identifier associated with each host or router *interface*
  - **address space** is **2^{32} (4 billion) hosts**
- *Each host on a **TCP/IP** internet is assigned a unique 32-bit internet address that is used in all communication with that host.*
- Initially deployed on Jan 1, 1983
- Still most commonly used version

- Expressed as four 8 bit Octets separated by **a period symbol** (Dotted decimal notation). e.g.,

10000111 00011001 00001101 10011000

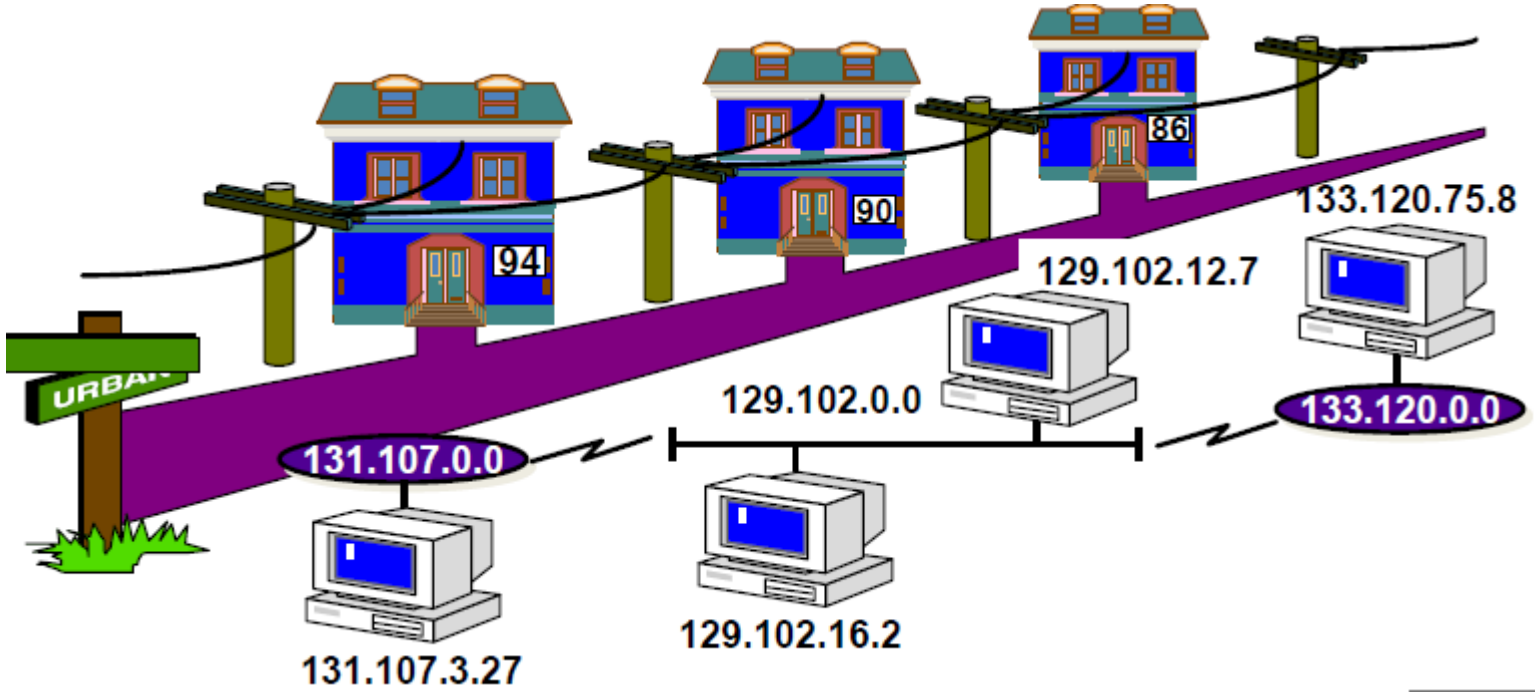
135.25.13.152

IP addressing (IPv6) : introduction

- Deployment started in 1999
- 128 bit numbers
- conventionally expressed using hexadecimal strings
  - FE80:CD00:0000:0CDE:1257:0000:211E:729C
- *Apparently Address space is* $\mathbf{2^{128}}$

Parts of an IP address

- **IP address** is divided into two parts
  - **Prefix** is the **Net ID** (like street address)
  - **Suffix** is the **Host ID** and it defines the node i.e., connection of a device to the network (like house or building number)



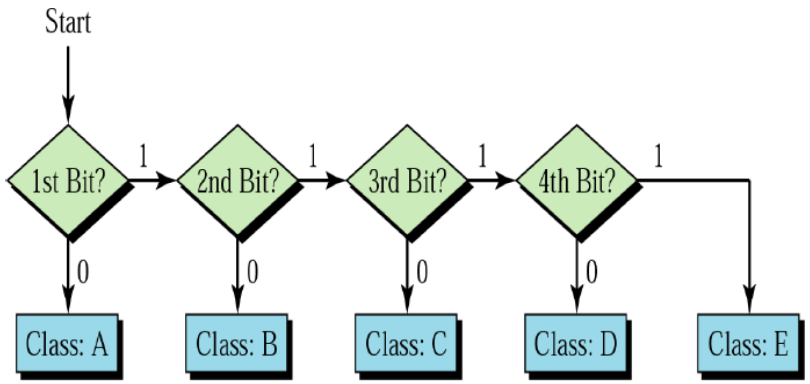
IPv4 classes

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-decimal notation



Hostid:

- cannot be all 0s (as all 0's, represents a **network** address)
- cannot be all 1s (as all 1's represent broadcast address)

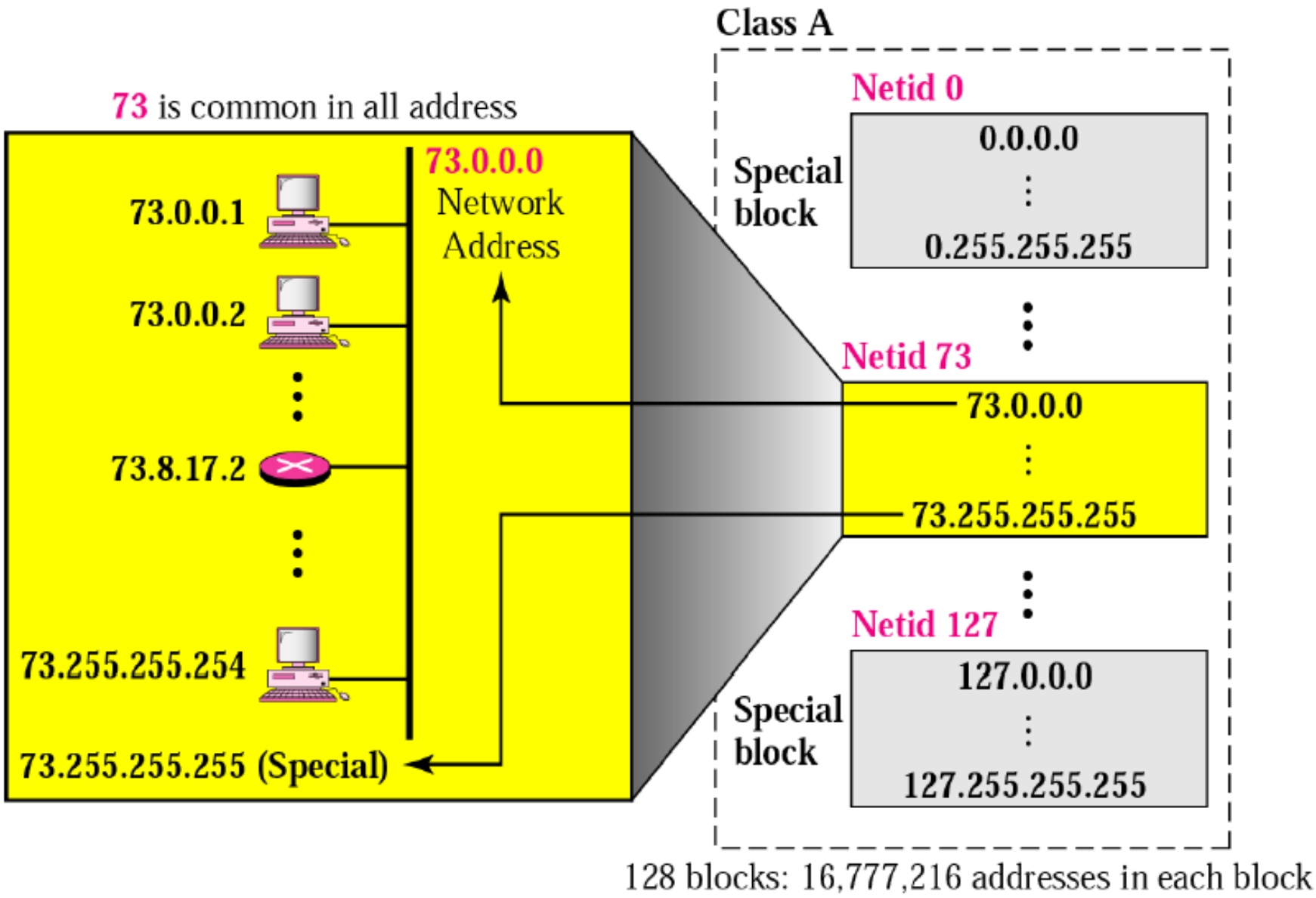
Class	Lowest Address	Highest Address
A	1.0.0.0	126.0.0.0
B	128.1.0.0	191.255.0.0
C	192.0.1.0	223.255.255.0
D	224.0.0.0	239.255.255.255
E	240.0.0.0	255.255.255.254

Class	*Number of Blocks*	*Block Size*	*Application*
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

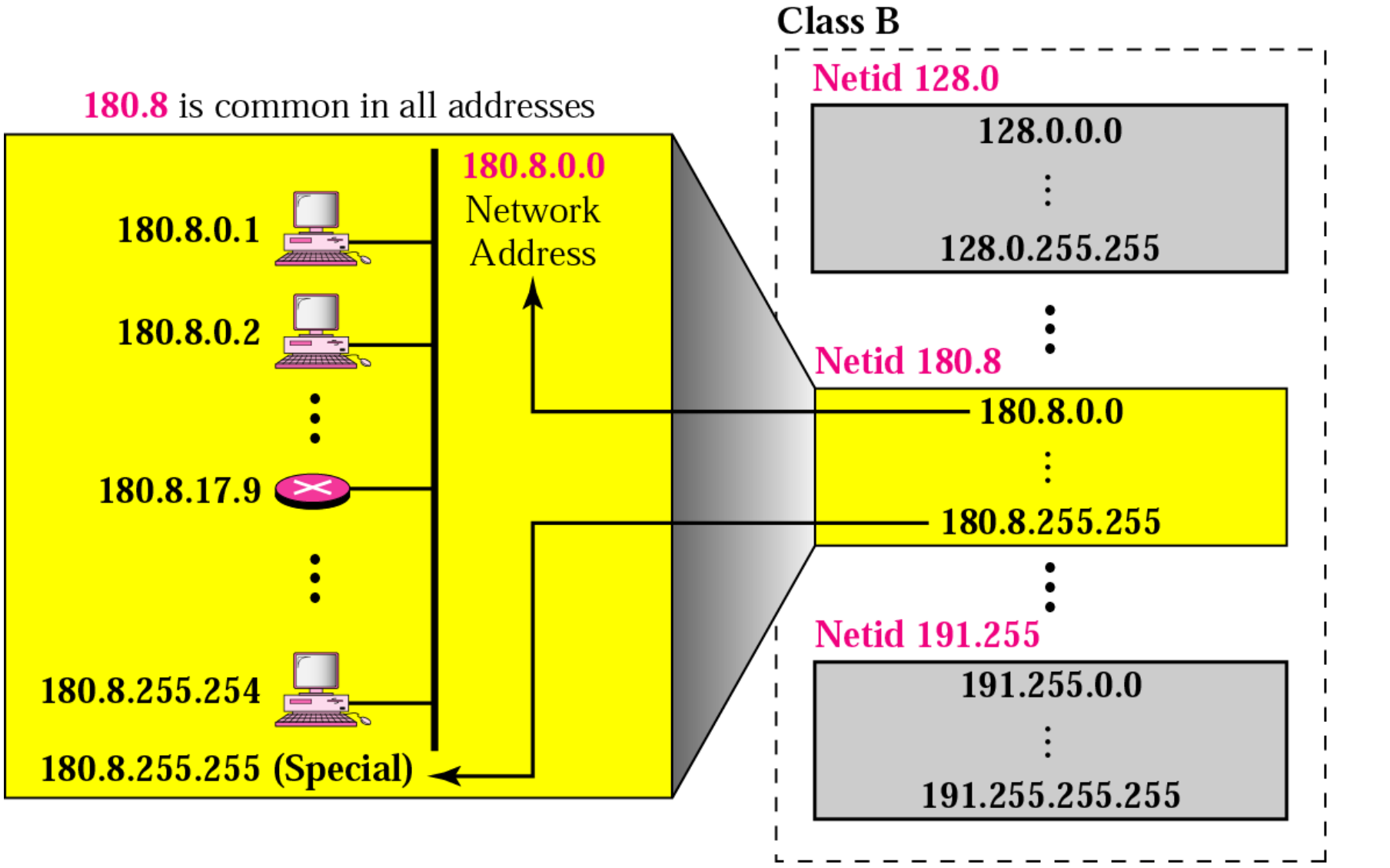
What is a host?

- A **host** is a device that has a network interface card (NIC) connected to a network.
- If a device has **two** network interfaces, it should be considered **two separate hosts.**
- Each host that is attached to a TCP/IP network must have a **unique IP address**

Class A Addresses

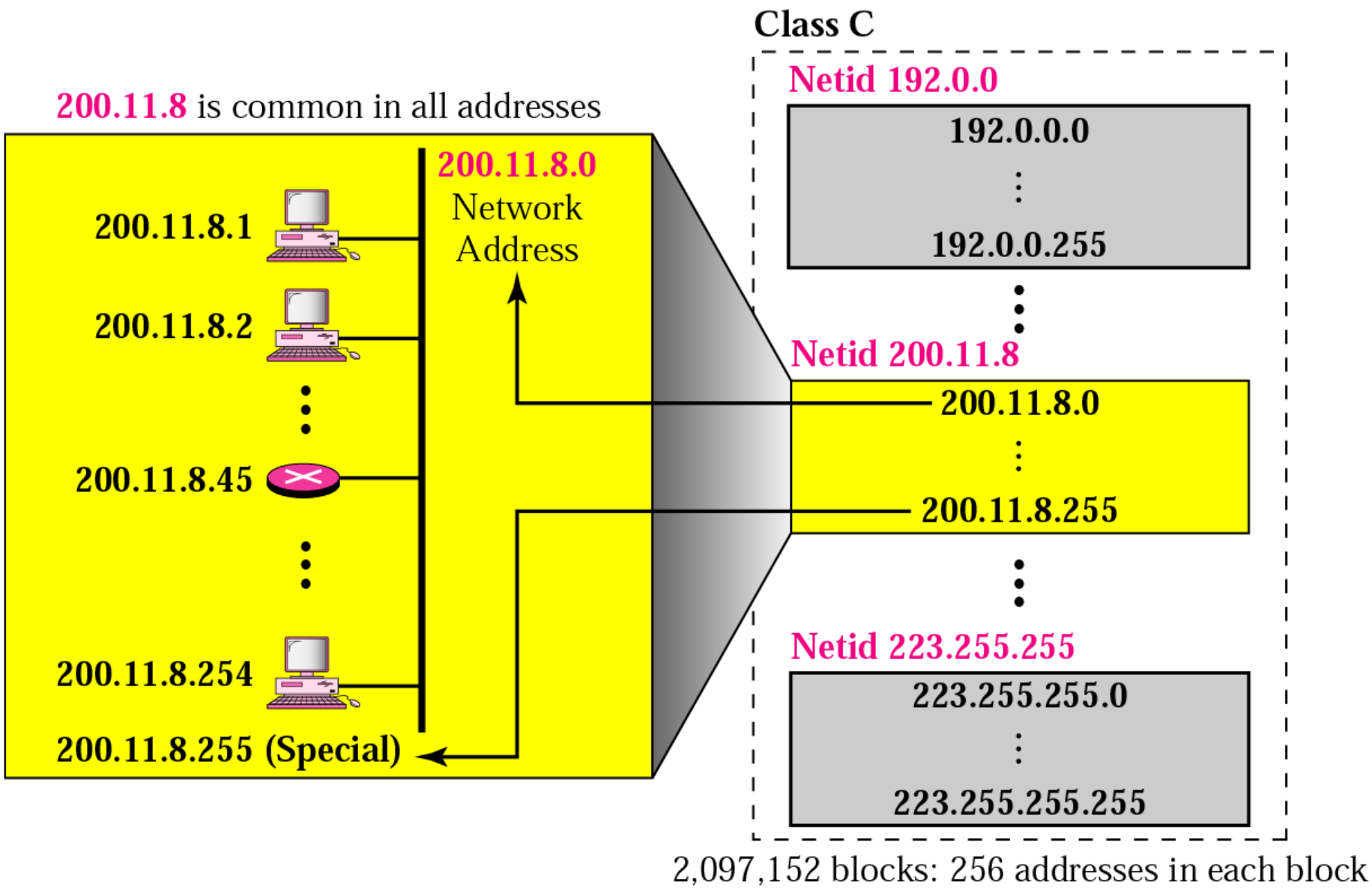


Class B Addresses



16384 blocks: 65536 addresses in each block

Class C Addresses



Class D and Class E Addresses

- **Class D addresses** (from 224.x.y.z to 239.x.y.z) are used for multicasting method of sending a single packet to multiple hosts

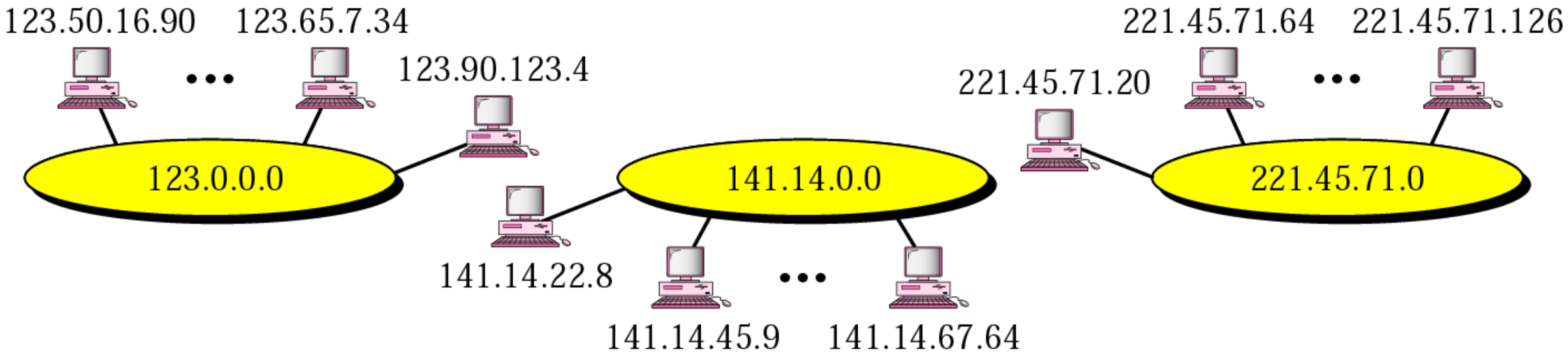
- **Class E addresses** (from 240.x.y.z to 255.x.y.z) experimental address range
- Not used in actual networks

Network Addresses

- **Network address**
  - Different from a netid
  - Contains **both netid and hostid** with **0s** for the **hostid**
- Defines the **organization network** (to connect to the rest of world).
- The **organization network** is **connected** to the **Internet** via a **router**.
- The router has **two addresses**.
  - One belongs to the **granted block**;
  - Other belongs to the network that is **at the other side of the router**.

Network Addresses

Netid	Hostid
Specific	All 0s



a. Class A b. Class B c. Class C

IP addressing: Subnet mask

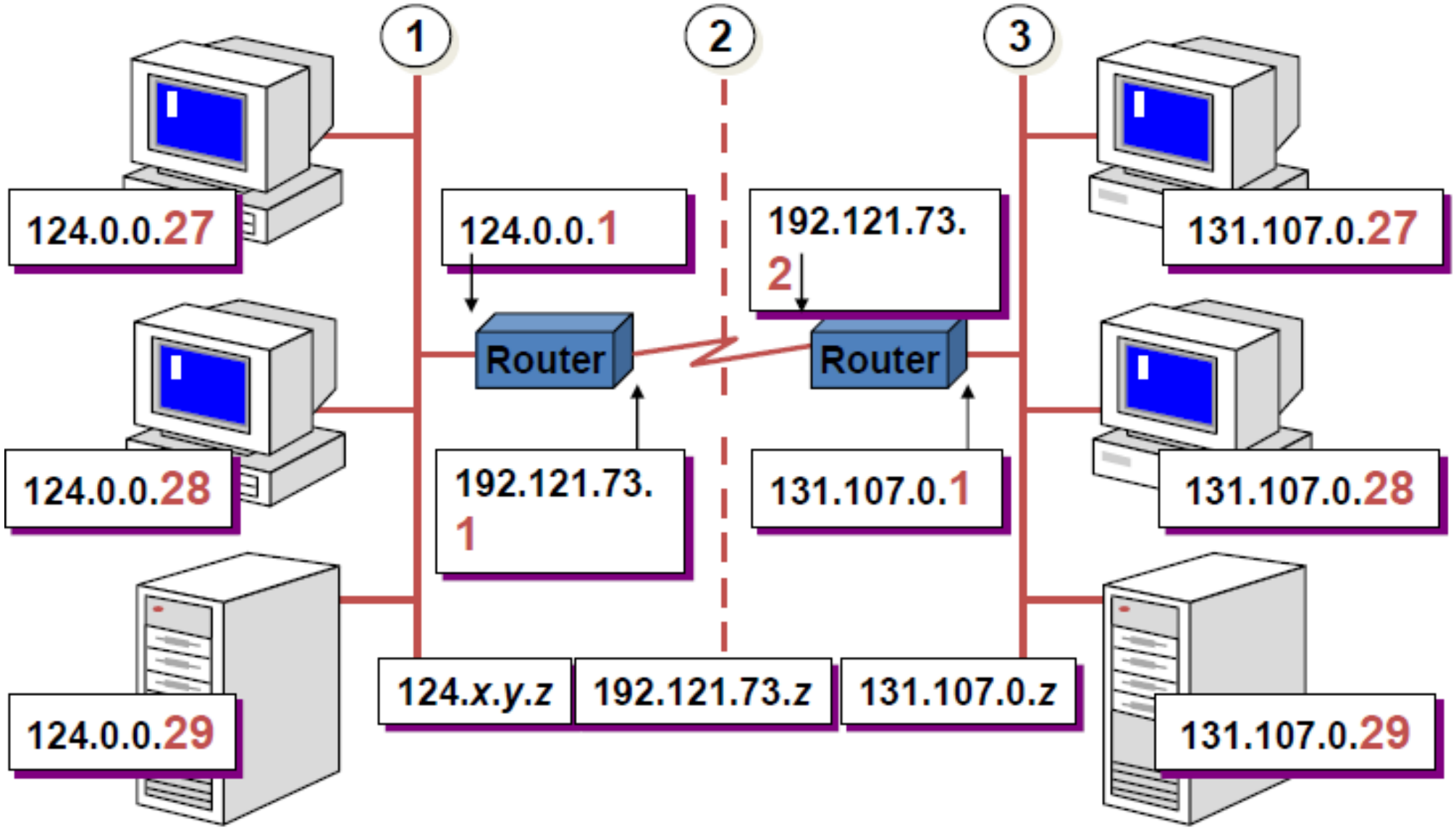
- **The purpose of subnet mask** is to identify which part is the **network part** and which part is the **host part**.
- A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s. In this way, the subnet mask separates the IP address into the network and host addresses.
- The “255” address is always assigned to a broadcast address, and the “0” address is always assigned to a network address. Neither can be assigned to hosts, as they are reserved for these special purposes.
- Class A: 255.0.0.0
- Class B: 255.255.0.0
- Class C: 255.255.255.0

Network Addresses Examples

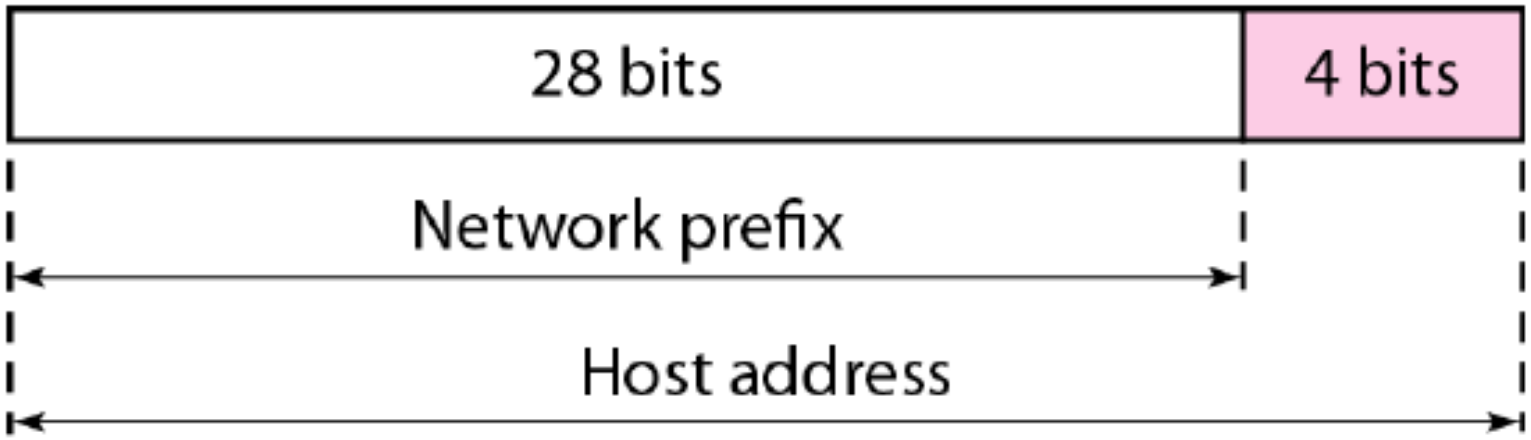
- **Question:** Given the address **23.56.7.91**, find the **network address**.
  - **Answer:**
  - **Network Mask (for Class A) is 255.0.0.0**
  - So, the network address is **23.0.0.0**.

- **Question:** Given the address **132.6.17.85**, find the **network address.**
  - **Answer:**
  - **Network Mask (for Class B) is 255.255.0.0**
  - Thus, the network address is **132.6.0.0.**

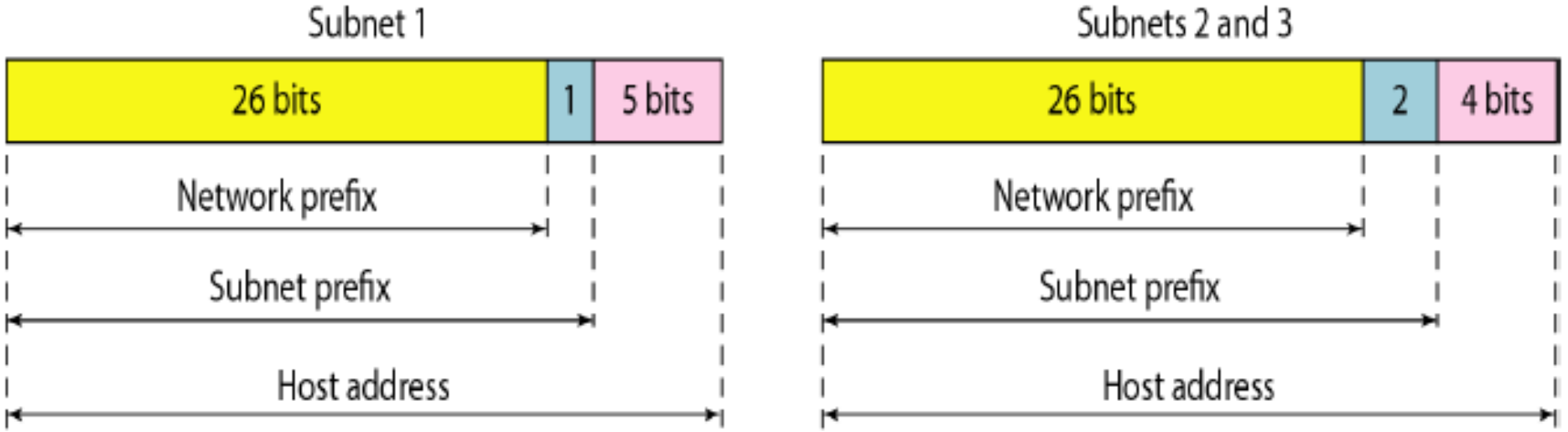
Assigning Host IDs



2-level hierarchy in IPv4 Addresses



Three-level hierarchy in an IPv4 address



Subnet and sub-netting

- **In sub-netting,** a class A or class B or class C block is divided into several subnets
- each subnet with larger prefix length than the original network).
- **For example,** divide the **class A** into **four subnets**, (take two bits from host id part in order to obtain subnets).then each subnet will have prefix length as 10

Class C sub-netting

- Consider class C network 192.168.10.0 (subnet mask is 255.255.255.0)
- If we include one bit from the host part to the network part, the subnet mask changes into 255.255.255.128 (for 1 bit sub-netting)
- The single bit can have two values in last octet, either 0 or 1
- 11000000.10101000.00001010.**0** | 0000000
  11111111.11111111.11111111.**1** | 0000000
- So the network 192.168.10.0 is divided into two networks with a single bit sub-netting, each network has 128 total addresses of which 126 are usable two are used in each subnet to represent the network address and broadcast address.

Class C 1-bit sub-netting

SN No.	Description	Binaries	Decimal
1	Network Address	11000000.10101000.00001010.00000000	192.168.10.0
	First usable address	11000000.10101000.00001010.00000001	192.168.10.1
	Last usable address	11000000.10101000.00001010.01111110	192.168.10.126
	Broadcast Address	11000000.10101000.00001010.01111111	192.168.10.127
2	Network Address	11000000.10101000.00001010.10000000	192.168.10.128
	First usable address	11000000.10101000.00001010.10000001	192.168.10.129
	Last usable address	11000000.10101000.00001010.11111110	192.168.10.254
	Broadcast Address	11000000.10101000.00001010.11111111	192.168.10.255

Class C 2-bit sub-netting

SN No.	Description	Binaries	Decimal
1	NA	11000000.10101000.00001010.00000000	192.168.10.0
	1st	11000000.10101000.00001010.00000001	192.168.10.1
	Last	11000000.10101000.00001010.00111110	192.168.10.62
	BA	11000000.10101000.00001010.00111111	192.168.10.63
2	NA	11000000.10101000.00001010.01000000	192.168.10.64
	1st	11000000.10101000.00001010.01000001	192.168.10.65
	Last	11000000.10101000.00001010.01111110	192.168.10.126
	BA	11000000.10101000.00001010.01111111	192.168.10.127
3	NA	11000000.10101000.00001010.10000000	192.168.10.128
	1st	11000000.10101000.00001010.10000001	192.168.10.129
	Last	11000000.10101000.00001010.10111110	192.168.10.190
	BA	11000000.10101000.00001010.10111111	192.168.10.191
4	NA	11000000.10101000.00001010.11000000	192.168.10.192
	1st	11000000.10101000.00001010.11000001	192.168.10.193
	Last	11000000.10101000.00001010.11111110	192.168.10.254
	BA	11000000.10101000.00001010.11111111	192.168.10.255

Class C 3-bit sub-netting

SN No.	Description	Binaries	Decimal
1	NA	11000000.10101000.00001010.00000000	192.168.10.0
	1st	11000000.10101000.00001010.00000001	192.168.10.1
	Last	11000000.10101000.00001010.00011110	192.168.10.30
	BA	11000000.10101000.00001010.00011111	192.168.10.31
2	NA	11000000.10101000.00001010.00100000	192.168.10.32
	1st	11000000.10101000.00001010.00100001	192.168.10.33
	Last	11000000.10101000.00001010.00111110	192.168.10.62
	BA	11000000.10101000.00001010.00111111	192.168.10.63
		.....	
8	NA	11000000.10101000.00001010.11100000	192.168.10.224
	1st	11000000.10101000.00001010.11100001	192.168.10.225
	Last	11000000.10101000.00001010.11111110	192.168.10.254
	BA	11000000.10101000.00001010.11111111	192.168.10.255

Class B 2-bit sub-netting

SN No.	Description	Binaries	Decimal
1	NA	10101100.00010000.00000000.00000000	172.16.0.0
	1st	10101100.00010000.00000000.00000001	172.16.0.1
	Last	10101100.00010000.00111111.11111110	172.16.63.254
	BA	10101100.00010000.00111111.11111111	172.16.63.255
2	NA	10101100.00010000.01000000.00000000	172.16.64.0
	1st	10101100.00010000.01000000.00000001	172.16.64.1
	Last	10101100.00010000.01111111.11111110	172.16.127.254
	BA	10101100.00010000.01111111.11111111	172.16.127.255
3	NA	10101100.00010000.10000000.00000000	172.16.128.0
	1st	10101100.00010000.10000000.00000001	172.16.128.1
	Last	10101100.00010000.10111111.11111110	172.16.191.254
	BA	10101100.00010000.10111111.11111111	172.16.191.255
4	NA	10101100.00010000.11000000.00000000	172.16.192.0
	1st	10101100.00010000.11000000.00000001	172.16.192.1
	Last	10101100.00010000.11111111.11111110	172.16.192.254
	BA	10101100.00010000.11111111.11111111	172.16.192.255

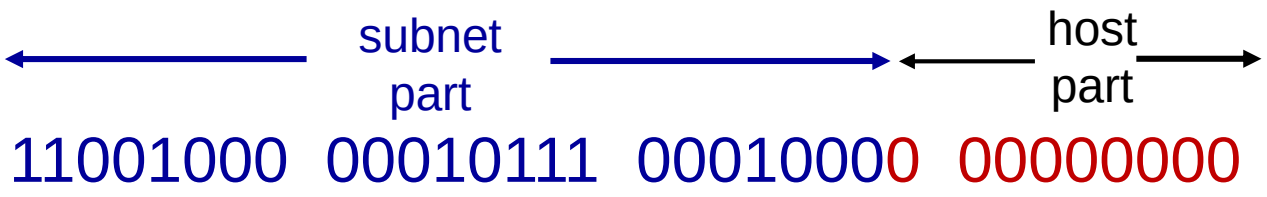
IP addressing: CIDR

- A system called **Classless Inter-Domain Routing** (CIDR), was developed as an alternative to traditional subnetting. You can add a specification in the IP address itself as to the number of significant bits that make up the routing or networking portion.
- For example, we could express the idea that the IP address 192.168.0.15 is associated with the netmask 255.255.255.0 by using the CIDR notation of 192.168.0.15/24. This means that the first 24 bits of the IP address given are considered significant for the network routing.

IP addressing: CIDR

CIDR: Classless InterDomain Routing (pronounced “cider”)

- subnet portion of address of arbitrary length
- address format: a.b.c.d/x, where x is # bits in subnet portion of address



200.23.16.0/23

Prefix Lengths

/n	Mask	/n	Mask	/n	Mask	/n	Mask
/1	128.0.0.0	/9	255.128.0.0	/17	255.255.128.0	/25	255.255.255.128
/2	192.0.0.0	/10	255.192.0.0	/18	255.255.192.0	/26	255.255.255.192
/3	224.0.0.0	/11	255.224.0.0	/19	255.255.224.0	/27	255.255.255.224
/4	240.0.0.0	/12	255.240.0.0	/20	255.255.240.0	/28	255.255.255.240
/5	248.0.0.0	/13	255.248.0.0	/21	255.255.248.0	/29	255.255.255.248
/6	252.0.0.0	/14	255.252.0.0	/22	255.255.252.0	/30	255.255.255.252
/7	254.0.0.0	/15	255.254.0.0	/23	255.255.254.0	/31	255.255.255.254
/8	255.0.0.0	/16	255.255.0.0	/24	255.255.255.0	/32	255.255.255.255

Subnet Example

Given the IP address 173.234.198.213/20.

(a) What is the subnet id?

(b) What is the usable hosts range? (starting and ending hosts IP addresses on the subnetwork).

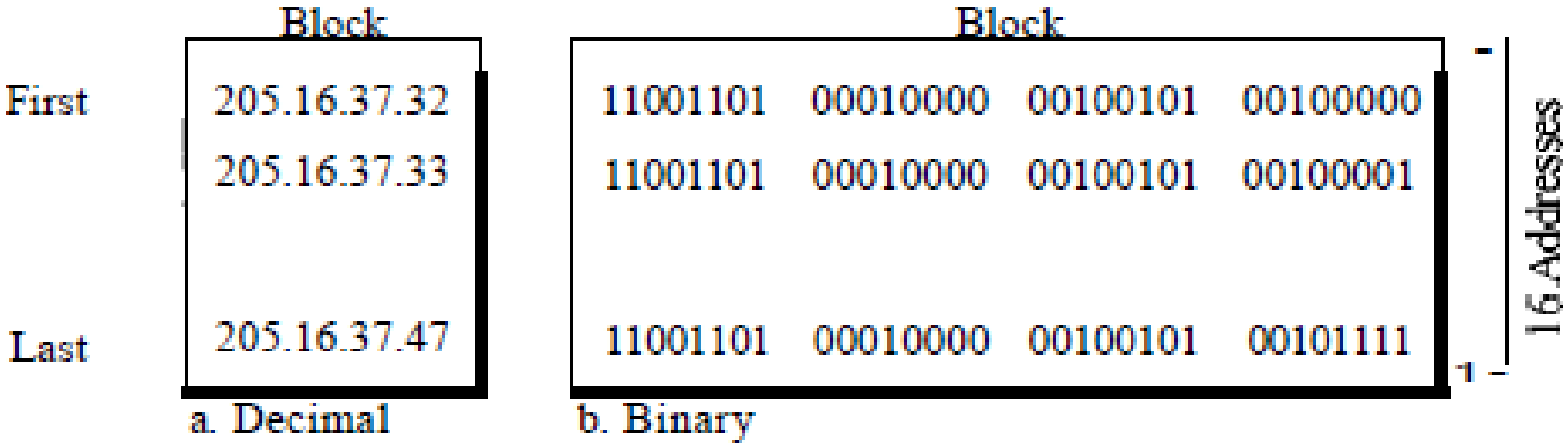
(a) Subnet ID: **173.234.192.0**

(b) Usable hosts range: **173.234.192.1** to **173.234.207.254**

Example

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28.

1. What is the first address in the block?
2. What is the last address in the block?
3. How many addresses this block contain?



Problem

- A company is assigned the CIDR block 192.168.0.0/24 for their internal network. They want to divide this block into four equal subnets for different departments. What is the new subnet mask for each subnet? List the first and last IP addresses for each subnet.
- **Solution:**
  - 255.255.255.192 (which corresponds to /26)
  - **Subnet 1:Network address: 192.168.0.0**
  - First usable IP: 192.168.0.1
  - Last usable IP: 192.168.0.62
  - Broadcast address: 192.168.0.63
  - **Subnet 2:Network address: 192.168.0.64**
  - First usable IP: 192.168.0.65
  - Last usable IP: 192.168.0.126
  - Broadcast address: 192.168.0.127
  - **Subnet 3:Network address: 192.168.0.128**
  - First usable IP: 192.168.0.129